

DEPARTMENT LIBRARY MANAGEMENT SYSTEM

A MINI-PROJECT REPORT

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Bachelor of Technology

In

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DECLARATION

We undersigned at this moment declare that the project report "**DEPARTMENT LIBRARY MANAGEMENT SYSTEM**", submitted for partial fulfilment of the requirements for the award of the degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bona fide work done by us under supervision of **Dr. Sujithra M S**, Head of Department, Department of CSE. This submission represents our ideas in our own words, and where ideas or words of others have been included, we have adequately and accurately cited and referenced the sources. We also declare that we have adhered to academic honesty and integrity ethics and have not misrepresented or fabricated any data, idea, fact, or source in our submission. We understand that any violation of the above will cause disciplinary action by the institute and/or the university and can also evoke penal action from the sources that have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for awarding any degree, diploma, or similar title of any other University.

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CERTIFICATE

This is to certify that the report entitled **“DEPARTMENT LIBRARY MANAGEMENT SYSTEM”** submitted by **Antony I Tom, Fayez Muhammed, Milan Martin, Sivaprasad S** to the APJ Abdul Kalam Technological University in partial fulfilment of the Computer Science and Engineering is a Bonafide record of the project work carried out by them under my/our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ABSTRACT

The Department Library Management System plays a critical role in the efficient organization and management of library resources, ensuring seamless operations and enhancing user experience. This project aims to design and implement a comprehensive Library Management System that streamlines the management of books, users, and transactions within a library. The system will include features such as providing administrative functionalities such as user registration, managing library resources and fine management. It Allows students to search for books, view book details, reserve and borrow books, renew borrowed items. It also allows organizing library resources into categories, subjects, authors, and topics for easy navigation and retrieval. The project provides a user-friendly interface, developed with modern tools and frameworks, enhances accessibility for both library staff and students. By automating routine tasks and reducing manual errors, the Library Management System optimizes resource utilization and facilitates efficient library operations. This project not only demonstrates the application of software engineering principles but also underscores the importance of digital solutions in modernizing traditional library systems.

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ABBREVIATIONS

API	Application Programming Interface
ISBN	International Standard Book Number
LMS	Library Management System
OPAC	Online Public Access Catalog
RFID	Radio Frequency Identification
SMS	Short Message Service
UI	User Interface

CHAPTER 1

INTRODUCTION

Library management has been central to human progress, adapting to the evolving needs of society over centuries. From primitive records on clay tablets to the advanced digital systems of today, libraries have continually developed new ways to organize and disseminate information. This evolution mirrors technological advancements, cultural shifts, and changing societal expectations, showcasing the enduring role of libraries as vital institutions of knowledge preservation.

The invention of the printing press in the 15th century marked a pivotal moment, making books more affordable and accessible. This technological advancement enabled libraries to expand their collections significantly, shifting the librarian's role from copying manuscripts to managing acquisitions and organization. National libraries also began to take shape, serving as repositories for cultural and historical artifacts. Figures like Sir Antonio Panizzi, who developed innovative cataloging methods and advocated for public access, exemplified the growing complexity of library management during this time.

By the 19th and 20th centuries, library management became increasingly professionalized. The establishment of organizations such as the American Library Association and the creation of formal library science programs helped standardize practices and equip librarians with the necessary skills to manage expanding collections. Libraries transitioned from static repositories to dynamic public institutions, offering educational programs, reference services, and outreach initiatives like traveling libraries. These efforts made knowledge more accessible to diverse communities, including those in rural or underserved areas.

A Library Management System has three basic functions to perform inside a library, referred by Sarma : 'Housekeeping Operations', 'Public Services', and 'Administrative Planning or Decision Making'. Housekeeping operations include acquisition of documents, technical processing like classification and cataloguing, circulation control, etc. Public services include online public access catalogue (OPAC), and Administrative planning or decision making includes formation of transaction policies, customisation of LMS etc. [1]

The Department Library Management System is a software solution developed to modernize and streamline the management of library resources within our department. Digitization of library helps to keep all the records secured and retrievable which saves the paperwork and made library

records easy to store. This project not only helps to store the library records but it also provides access to the library management staffs and students to check the information related to the books. Traditionally, maintaining library records involved manual processes, which are not only time-consuming but also prone to errors like misplaced books, duplicate records, or delayed updates. This system replaces these outdated methods with an efficient, automated approach, allowing for smooth management of books, student and faculty records, and transactions. With features like adding, updating, and searching for books, it ensures that library operations are quicker and more accurate, reducing the workload on staff while improving overall organization. In addition to basic management functions, the system introduces tracking for book issuance and returns, ensuring users have timely access to the resources they need while avoiding penalties for overdue books. Inventory management is also simplified, allowing staff to monitor book availability and replenish resources when needed. The user-friendly interface makes it easy for anyone to navigate the system, whether they are library staff, students, or faculty. By improving efficiency and minimizing errors, this system not only enhances the library's day-to-day operations but also promotes a better experience for everyone, creating a well-organized and accessible resource hub for the department.

1.1 OBJECTIVE

- The primary objective of the Library Management System project is to design and develop an efficient and secure software application that automates and streamlines the operations of a library. .
- The system aims to enhance accessibility and convenience for users by providing advanced search functionalities and minimizing manual errors and administrative workload through automation of routine tasks.
- Developing an intuitive and user-friendly interface for library staff and patrons to facilitate easy navigation
- bridges the gap between traditional library operations and modern digital solutions, promoting efficient resource management and improved user experience.
- adopts an organized and systematic approach to library resource management.

1.2 SCOPE

The Library Management System project has a broad and impactful scope, aiming to digitize and optimize library operations for improved efficiency and user satisfaction. The system will be designed to cater to the needs of both library administrators and users. For administrators, it provides tools to manage book inventory, monitor borrowing trends, handle overdue items, and generate reports for better decision-making. For users, the system offers a streamlined experience with features such as book search by title, author, or genre, real-time availability updates, and seamless borrowing and returning processes. The project has potential for future enhancements, such as integrating e-books, online reservation systems, and support for mobile applications, thereby making the library accessible anytime and anywhere. This project serves as a practical implementation of software engineering concepts and database management systems, addressing real-world challenges faced by traditional libraries and paving the way for modern and efficient library operations.

CHAPTER 2

LITERATURE SURVEY

2.1 EVOLUTION OF LIBRARY MANAGEMENT SYSTEMS: PAST, PRESENT AND FUTURE

Past: The development of library management dates back to ancient civilizations, where libraries were primarily focused on organizing and preserving knowledge. Early examples from Mesopotamia and Egypt featured libraries as places for spiritual and intellectual enrichment. As time progressed, especially in Ancient Greece and Rome, libraries developed systems for organization and management. The invention of the printing press in the 15th century further influenced the growth of libraries, making books more accessible and shifting the role of librarians.

Modern: library management has evolved into a complex, strategic discipline influenced by technological advancements and changing user expectations. Modern libraries have transitioned from being passive repositories to proactive service providers, adopting business management concepts such as strategic planning, customer service, and cost analysis. Professional associations like the American Library Association (ALA) and the Library Leadership and Management Association (LLAMA) provide support and resources for library managers, while library education has become more specialized, with degrees like MLS and MLIS becoming standard. The integration of information technology (IT) has transformed library operations, with the automation of business functions, the creation of online catalogs, and the adoption of electronic databases. However, libraries now face the challenge of balancing physical and digital collections, as well as addressing issues like copyright, digital preservation, and interoperability.

Future: Looking ahead, library management faces challenges and opportunities brought about by digital transformation. Libraries are increasingly focused on creating digital collections, collaborating globally, and embracing new technologies to meet user expectations. However, they must navigate issues like digital preservation, copyright, and system interoperability. The future of libraries will require strong leadership, strategic innovation, and a balance between maintaining physical collections and developing new digital resources to ensure continued relevance in an increasingly digital world.[2]

2.2 ONLINE LIBRARY MANAGEMENT SYSTEM FOR ENHANCED ACCESSIBILITY AND EFFICIENCY

Online Library Management System is an Automated Library System that handles the various functions of the library. It provides a complete solution to the library management software. The online Library Management System is classified into two parts: Bar Code System and RFID System.

Library plays an important role in all schools and colleges, no educational institution can exist without Library Administration Software. It is an important part of every school and college and it helps the librarian to keep records of available books as well as issued books. Library Management System software helps in different ways by providing students the facility to learn, gather resources, promote group learning and improve knowledge and skills.

The demand for Library Automation Software in India is increasing day by day with the advancement of the system. As we all are living in the digital age, everyone prefers quick and accurate service. Almost all schools and colleges prefer the Library Management System due to the organized and systematic management of the database. College Library Software helps the students to prepare their projects easily and in less time. It also helps in preparing notes and assignments.

Library Management System helps the students in better learning by providing quick access to the library system. With Advance Library Software they can easily find books, catalogs, magazines of their interest. The library is a great place to study and make notes which ultimately help in scoring a good score.[\[3\]](#)

2.3 FUNCTIONAL CAPABILITIES OF LIBRARY MANAGEMENT SYSTEMS

Library Management System has been intended to automate, oversee and care for the general handling of even enormous scale libraries. This product is fit for overseeing Book Issues, Returns, Calculating/Managing various data, creating different Reports for Record-Keeping and Review purposes, as per end client prerequisites.

it also needs to keep a thorough database of the members, the Management System stores each user's name, ID, and password for aiding in determining the member's history. The software is user-friendly and improves the effectiveness of the librarian and library administration.

A modern integrated library management system (LMS). Can be scalable to Windows, Linux and Mac OS platform, Full catalog, circulation and acquisitions system for library stock management, Simple, clear search interface for all users, Multilingual and multi-user support, Export and import records, Easy way to enter new books, Keep record of complete info of a book like title, author name, publisher name etc, Easy way to check-in and check-out, It is user-friendly software, Keep record of different categories, Classify the books subject wise, Easy to maintain database,[4]

2.4 KEY CHALLENGES AND LIMITATIONS IN LIBRARY MANAGEMENT SYSTEMS

Library management systems (LMS) face several challenges related to user experience. A complex or non-intuitive user interface (UI) can make it difficult for both librarians and patrons to navigate the system effectively. Furthermore, performance issues, especially in large libraries with extensive collections, can hinder the system's efficiency, particularly during peak hours. Many systems also struggle with scalability, making it difficult to handle growing collections and user bases. Effective search functionality is another key issue, with many systems lacking intuitive search options, hindering users' ability to quickly locate materials. Poor reporting tools also present problems, as limited or poorly designed reporting capabilities prevent library staff from making data-driven decisions.

The integration capabilities of LMS are often limited, restricting the ability to connect the system with other important tools like accounting software or e-learning platforms. Additionally, data migration can become problematic when transitioning from one system to another, with poorly designed migration tools leading to data loss or corruption. Many systems also fail to provide sufficient automation for routine tasks such as sending overdue notices or performing inventory checks, adding unnecessary manual work for library staff. Mobile accessibility is another area where many LMS solutions fall short, making it difficult for patrons to access services via smartphones or tablets. Moreover, the inflexibility of lending rules in many systems limits customization for libraries with unique policies.

Security vulnerabilities remain a major concern for LMS, as insufficient protection can lead to unauthorized access to user data and library records. Limited customization options also hinder libraries from tailoring the system to meet their specific operational requirements, leading to inefficiency. Compatibility issues across different operating systems and devices can also limit

accessibility, forcing users to deal with inconsistent experiences across platforms. Furthermore, a lack of proper user training from LMS providers can lead to inefficiencies and user frustration, as libraries are left to figure out the system on their own. Inadequate technical support exacerbates this issue, leaving libraries without timely assistance when problems arise.

The cost of upgrading LMS software is another challenge, as many providers charge high fees for upgrades, creating financial and operational barriers for libraries. Additionally, inadequate backup and recovery options pose a significant risk to library data, especially in the event of hardware failures or cyberattacks. Licensing restrictions can further complicate library operations by limiting the number of users or installations allowed under the license, leading to additional costs as libraries grow. Many LMS solutions also struggle to manage digital resources, such as e-books and audiobooks, which are increasingly important in modern libraries. Finally, issues with user account management, such as complicated registration processes and password reset problems, frustrate both staff and patrons.[5]

2.5 SUMMARY OF SURVEY

The evolution of Library Management Systems (LMS) reflects a journey from ancient knowledge repositories to today's digitally driven platforms. Historically, libraries in Mesopotamia, Egypt, Greece, and Rome prioritized preservation and organization, with the invention of the printing press in the 15th century marking a pivotal shift toward accessibility and public service . In the modern era, LMS has transitioned into a strategic discipline, integrating automation (online catalogs, electronic databases) and business principles like cost analysis and user-centric services. This evolution has been accelerated by the rise of online LMS, which uses technologies to automate core functions such as inventory tracking, book issuance, and returns. These systems are indispensable in educational institutions, enabling remote access, reducing manual errors, and supporting student productivity through quick resource discovery and collaborative tools . The digital transformation of libraries underscores the need to balance physical collections with emerging technologies while addressing challenges like interoperability and digital preservation. To build a future-ready LMS, essential features such as automated workflows (fine calculation, report generation), intuitive interfaces (multilingual search, cross-platform compatibility), and robust catalog management (subject-wise classification, centralized member databases) must be prioritized. However, persistent challenges—including poor user experience (complex UI, limited mobile access), scalability constraints, and high operational costs—demand proactive

solutions. Additionally, cost barriers can be mitigated through modular or open-source LMS models, while comprehensive training and technical support ensure smooth adoption. By synthesizing historical insights, modern technological advancements, and user-centric design principles, libraries can overcome existing limitations and evolve into agile, hybrid institutions that seamlessly blend physical and digital resources, ensuring relevance in an increasingly interconnected, digital-first world.

2.6 PROBLEM DEFINITION

Modern libraries are transitioning from traditional, manual systems to digital platforms to align with the accessibility and efficiency demands of the digital age. However, existing Library Management Systems (LMS) face critical challenges that impede their effectiveness. Many LMS platforms suffer from outdated user experiences, characterized by complex interfaces, inefficient search tools, which frustrate users and reduce operational productivity. Furthermore, technical limitations such as poor scalability, and performance lags during peak usage hinder their ability to adapt to growing collections and user bases. Reliance on manual processes for tasks like inventory management and overdue reminders increases administrative burdens and errors. Additionally, the rise of hybrid collections (physical and digital resources) has highlighted the inadequacy of many LMS in managing e-books, and other digital media, limiting their relevance in today's tech-driven environment. These challenges collectively undermine libraries' roles as dynamic knowledge hubs, necessitating an innovative LMS solution that prioritizes intuitive design, scalability, cost efficiency, and automation while seamlessly bridging physical and digital resource management. Addressing these gaps is crucial to enhancing operational efficiency, user satisfaction, and the long-term relevance of libraries as vital educational and community institutions.

CHAPTER 3

SYSTEM REQUIREMENT

3.1 SOFTWARE REQUIREMENTS SPECIFICATION

The Software Requirements Specification (SRS) is a foundational document in the software development lifecycle, serving as a blueprint for the development, testing, and maintenance of a software system. It provides a detailed description of the system's intended behavior, outlining both the functional and non-functional requirements that define what the system must accomplish and how it should perform. The functional requirements describe the system's features, such as user interactions, inputs, outputs, and processes, while non-functional requirements focus on performance metrics, security, reliability, and usability. An SRS not only ensures that all stakeholders—developers, testers, project managers, and clients—share a common understanding of the system's scope and objectives but also helps in minimizing misunderstandings and ambiguities. Additionally, it acts as a reference point throughout the entire software development process, supporting design decisions, providing a basis for quality assurance, and guiding the future evolution or maintenance of the system. By documenting assumptions, constraints, and dependencies, the SRS helps anticipate potential risks and challenges. Ultimately, the SRS is an essential tool for ensuring that the final software product meets the user's needs and adheres to the agreed-upon specifications.

3.1.1 INTRODUCTION

The purpose of the Library Management System (LMS) is to provide an efficient and user-friendly software solution for managing the day-to-day operations of a library. The system aims to automate and streamline processes such as book cataloging, user registration, borrowing and returning books. Ultimately, the system will aim to simplify the management of library resources while providing a seamless experience for all users.

The scope of this project includes the following key functionalities:

- Registration, authentication, and role-based access control for users
- Adding, updating, and removing books from the library catalog.
- Enabling users to search the catalog by various criteria like title, author.

- Recording all transactions, including borrowed and returned books and user activity.
- Providing insights to help manage the library effectively.

3.1.2 OVERALL DESCRIPTION

The Library Management System (LMS) is a standalone application designed to automate and enhance the core functionalities of library operations. It will serve as a centralized system for managing library resources, including books, users, and transactions. The system will be deployed on a web-based platform, making it accessible to both library staff and members from any device with internet access. The LMS will integrate seamlessly with a backend database to store and manage data on books, users, and transactions.

The Product Functions are as follows :

- Database Management: Maintain a comprehensive database of library resources, including information about new acquisitions, borrowed items, and due dates.
- Automated Processes: Streamline tasks such as book indexing, cataloging, reservations, and notifications to users.
- Categorization: Organize library resources into categories, subjects, authors, and topics for easy navigation and retrieval.
- User Management: Manage library memberships, issue library cards, and handle user requests and inquiries.
- Admin Panel: Provides administrative functionalities such as user registration, managing library resources, generating reports, and maintaining records.
- Student Panel: Allows students to search for books, view book details, reserve and borrow books, renew borrowed items, and analyze their transaction history.
- Faculty Staff Panel: Similar to the student panel, tailored for faculty and staff members, enabling them to access library resources and perform relevant tasks.

The User Characteristics are as follows:

- General Users: General users are the members of the library who will use the system to search for books, borrow and return books, and manage their personal account information. Their primary interaction with the system will be related to managing their borrowing activity and viewing the library's catalog.

- **Admin Users:** Admin users, typically librarians or library staff, will have full access to the system's administrative functions. They will be responsible for managing the library's book catalog, including adding, updating, or removing books. Admin users will also manage user registrations, define roles, and generate reports on library activity.

3.1.3 FUNCTIONAL REQUIREMENTS

USER MANAGEMENT:

- **User Registration:** Librarians can create, update, and deactivate patron accounts. Mandatory fields: name, contact info, and library card number.
- **Role-Based Access Control:** Define roles (Admin, Patron). Admins manage system settings, Librarians handle transactions, Patrons access self-service features.
- **Patron Self-Service:** Patrons can view checkouts, holds, fines, update contact info, and renew items via a secure portal.

CATALOG MANAGEMENT

- **Add/Edit Catalog Entries:** Librarians can add/update items (books, e-books, DVDs) with details (title, author, ISBN, format, copies).
- **Search and Browse:** Users can search by title, author, genre, or keyword. Filters for availability, format, and publication year.
- **Inventory Tracking:** Track item status (available, checked out, lost) and manage multiple copies.

CIRCULATION MANAGEMENT

- **Check-In Items:** Automatically update status, calculate fines if overdue, and notify reserved patrons.
- **Renewals:** Allow renewals (up to 3 times) if no holds exist.

RESERVATIONS & HOLDS

- **Place Holds:** Patrons can reserve unavailable items.

- Hold Queue Management: Librarians can view/modify hold queues and manually assign items.

FINES & FEES

- Automatic Fines Calculation: Overdue fines will be collected on a daily basis, with the option for waivers subject to approval.

REPORTING & ANALYTICS

- Generate Reports: Overdue items, checkout history, popular items, fines collected. Export to PDF/Excel.
- Real-Time Dashboard: Display daily checkouts, active holds, and overdue items for librarians.

NOTIFICATIONS

- Automated Alerts: Email/SMS for due dates, overdue items, holds availability, and account changes.

SYSTEM CONFIGURATION

- Manage Policies: Admins set loan periods (e.g., 14 days), fine rates, max checkouts (e.g., 10 items), and hold expiration.

SECURITY & COMPLIANCE

- Authentication: Secure login with password policies, and session timeout (15 minutes).

BACKUP & RECOVERY

- Data Backup: Schedule daily backups and allow manual backups. Restore data from backups.

ADDITIONAL FEATURES

- No-Due Calculation: Generates no due report based on book yet to be returned.

3.1.4 NON-FUNCTIONAL REQUIREMENTS

USABILITY

- The system shall have an intuitive and user-friendly interface suitable for users with basic computer literacy.

COMPATIBILITY

- The system shall be compatible with major web browsers.(e.g., Chrome, Firefox, Edge, Safari).

PORTABILITY

- The system shall be deployable on multiple operating systems (e.g., Windows, Linux, macOS).

AVAILABILITY

- The system will be available whenever necessary.

3.2 HARDWARE AND SOFTWARE REQUIREMENTS

3.2.1 HARDWARE REQUIREMENTS

- Processor: Intel i3/AMD Ryzen 3 (minimum), Intel i5 / AMD Ryzen (recommended)
- RAM: 4GB (minimum), 8GB or higher (recommended)
- Storage: 10GB HDD (minimum), 20GB SSD (recommended)
- Internet: Stable broadband connection (5Mbps minimum), High-speed (10Mbps+ recommended)

3.2.2 SOFTWARE REQUIREMENTS

- Supported OS: Windows 10/11, macOS 10.14+, Linux (Ubuntu 20.04+), Chrome OS
- Browser Compatibility: Latest versions of Chrome, Firefox, Edge, Safari
- Frontend: React with TypeScript, Tailwind CSS, Shadcn UI
- Backend: Node.js with Express.js

- Database: MongoDB with Mongoose ODM
- Development Tools: VS Code, MongoDB Compass, Postman

CHAPTER 4

SOFTWARE DESIGN

4.1 SYSTEM ARCHITECTURE

The Library Management System follows a 3-tier architecture [6] to separate concerns and ensure modularity and is shown in Fig 4.1.1

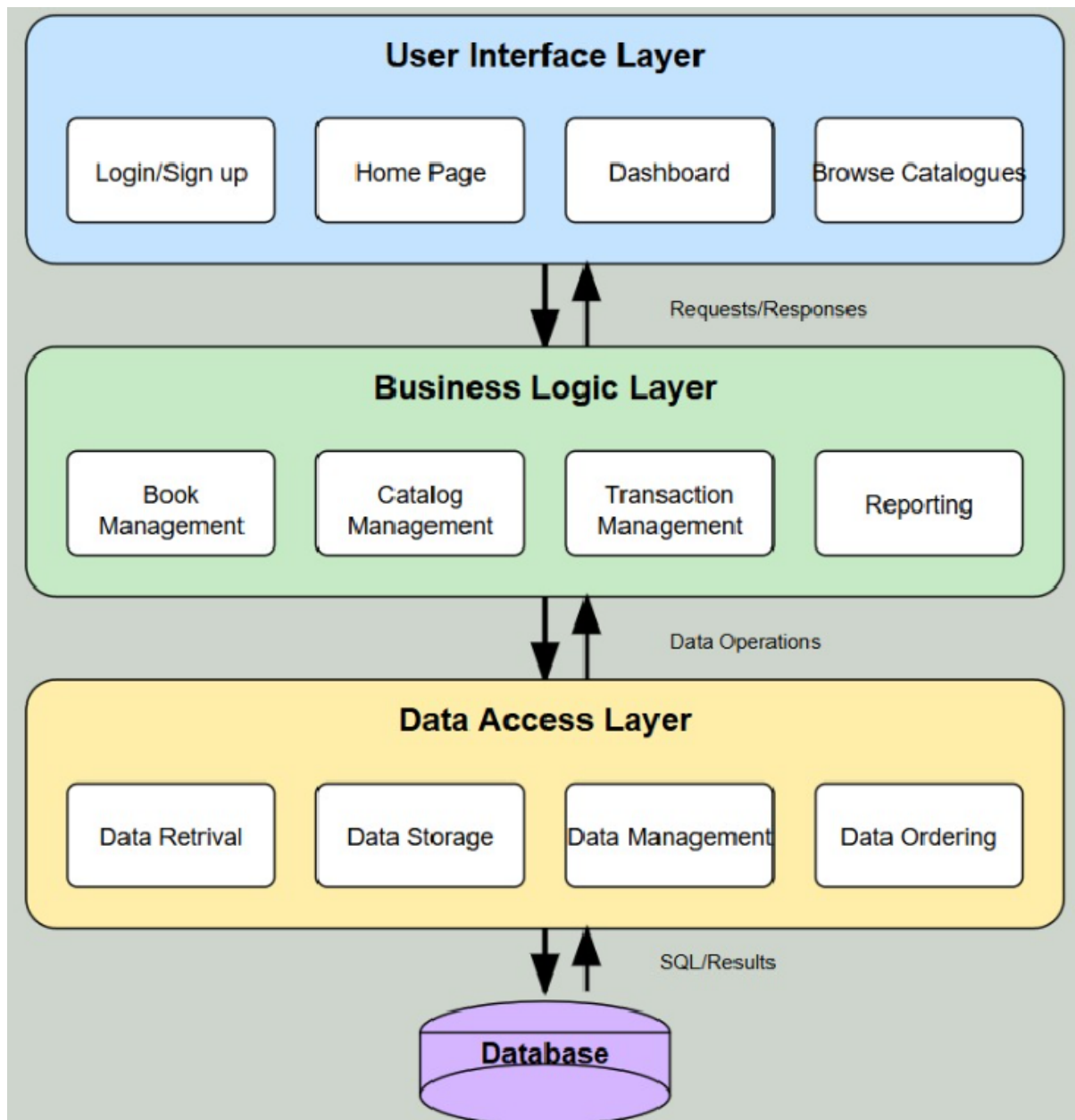


Fig 4.1.1 Three Tier architecture of library management system

4.1.1 User Interface Layer [Front-end]

This layer is responsible for interacting with the end-users of the system. It includes An interface for library staff and patrons to manage books, members, and transactions. The application is composed of a series of interactive web pages. Through this layer, users can directly enter the system to complete the required functions. According to the user category, different users would log in to the system with different identities. There are three types of this system, namely, students, teachers and librarian. Their permissions differ generally from low to high Teachers and students can only borrow, read and consult library archives. Differences exist between teachers and students in the scope and right of use. The librarian can sort out and optimize various resources of library management system, and gradually improve the level of user interface. The user layer is usually logged in as a browser, which is extremely convenient to use and has a friendly interface.

4.1.2 Business logic layer [Back-end]

This layer contains the core business logic of the system and is responsible for processing requests from the UI layer. It acts as the controller of the system. Handle incoming requests, process them, and send responses back to the UI layer. It implement the core functionalities such as book management, catalog management, transaction processing, and reporting. The Application layer is located between the user interface layer and the data layer of the hierarchical model. Application layer is mainly used to realize the business functions of the management system, which is an important layer in the hierarchical model. This system comprised several modules, such as user registration, user login, literature resource search, literature resource reservation and so on.

4.1.3 Data Access Layer [Data Storage & Management]

This layer is responsible for data storage, retrieval, and management. Stores all the data related to books, members, and transactions. The data layer has access to external databases. It is the bottom layer of the whole system. By interacting with the database, we can complete operations such as query, insert, delete and modify. The data access layer is mainly responsible for data extraction and data editing of the application layer. The system includes user information, book content data, etc. The user's basic data comprises the basic data of general users and

librarians. The content data of the website is mainly about book ordering information, borrowing information, etc.

4.2 MODULE DESCRIPTION

The Department Library Management System is a comprehensive digital platform designed to streamline library operations, serving both administrators and users (students and staff) with robust, user-friendly features.

4.2.1 ADMINISTRATOR MODULE

The Administrator Module is designed for librarians or system administrators to efficiently manage books, users, and reports within the library management system. This module provides complete control over the library's inventory, book issue/return process, and fine management. Administrators ensure that books are properly issued, returned, and tracked while monitoring overdue books and fines. In book management, administrators can add, update, or remove books from the system, storing essential details such as title, author, ISBN, category, and availability status in the book database.

The book issue and return process allows librarians to issue books to students and staff, recording borrower details and assigning due dates based on library policies. Upon return, the system updates the loan record and changes the book's availability status. If a book is overdue, fines are automatically calculated, and notifications are sent to users. Additionally, the reservation system allows users to reserve unavailable books, with notifications sent when the book is returned. Reservations expire after a fixed time if not collected.

User management enables administrators to add, update, and remove users, categorizing them as students, who have a limited borrowing period and fewer books, or staff, who have a longer borrowing period and can borrow more books. Role-based permissions are enforced to regulate user privileges. Furthermore, the report generation feature provides insights into issued and overdue books, pending reservations, and collected or unpaid fines, helping administrators analyze library usage and book demand effectively.

4.2.2 USER MODULE

The User Module is designed to facilitate students and staff in efficiently managing their library activities, including searching for books, making reservations, borrowing, and returning books.

To ensure secure access, the module incorporates an authentication system where users must log in before utilizing any library services. New users can register by providing their personal details, which helps in creating a personalized account for tracking their library activities. This authentication ensures that only authorized users can access the system and borrow the books they want.

One of the key features of this module is the book search and reservation system. Users can search for books based on various criteria such as title, author, ISBN, or category, making it easier to find the desired books. If a book is available, users can borrow it according to the borrowing policies set by the library. However, if the book is currently unavailable, users have the option to reserve it. Once the book is returned, the system notifies the user, ensuring they get an opportunity to borrow it. Additionally, the borrowing and return management system helps users keep track of their borrowed books, view due dates, and request renewals before the due date. Borrowing limits are enforced to ensure fair access to books, and if a book is overdue, the system automatically calculates fines to encourage timely returns.

Users also have access to profile management features, allowing them to update their personal details, such as name, email, and contact information. This ensures that their information remains up to date, facilitating better communication between the library and the users. Furthermore, users can review their borrowing history, helping them track past transactions and stay informed about their reading habits. By providing these comprehensive features, the User Module enhances the overall library experience, making it more efficient and user-friendly.

4.3 USE CASE DIAGRAM

The Use Case Diagram visually represents the interactions between users and the system showcasing different functionalities available to them.

The various actors are:

1. Librarian:

The librarian is responsible for managing the library system, including user and book management as shown in Fig 4.3.1. Primary Responsibilities:

- Managing users (adding, deleting, and modifying users).
- Issuing and returning books.
- Adding and removing books.

- Managing book records (adding, editing, and deleting book details).

2. Users (Students & Staff)

Users are the library members who can access library services as shown in Fig 4.3.2. They are divided into two categories: Students and Staff. Primary Responsibilities:

- Searching, reserving, returning and viewing books.
- Paying fines if applicable.
- Accessing e-books.
- Changing credentials (such as passwords).
- Registering as new users by filling up a registration form.

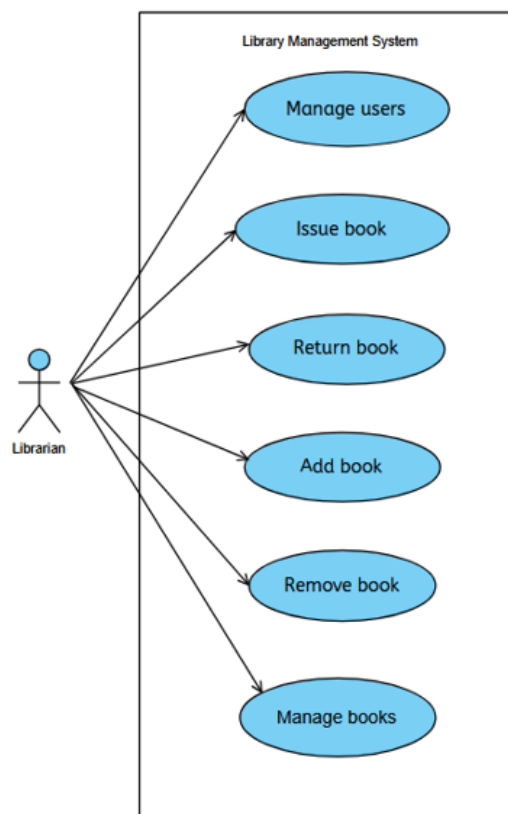


Fig 4.3.1 Use Case for Librarian

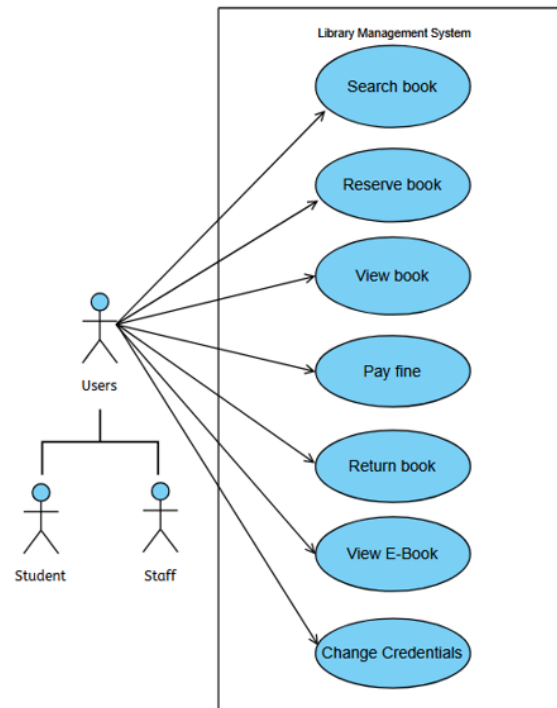


Fig 4.3.2 Use Case for User

4.4 DATA FLOW DIAGRAM

The Data Flow Diagram (DFD) is a structured analysis and design method. It is a visual tool to depict logic models and expresses data transformation in a system.

4.4.1 LEVEL-0 DFD

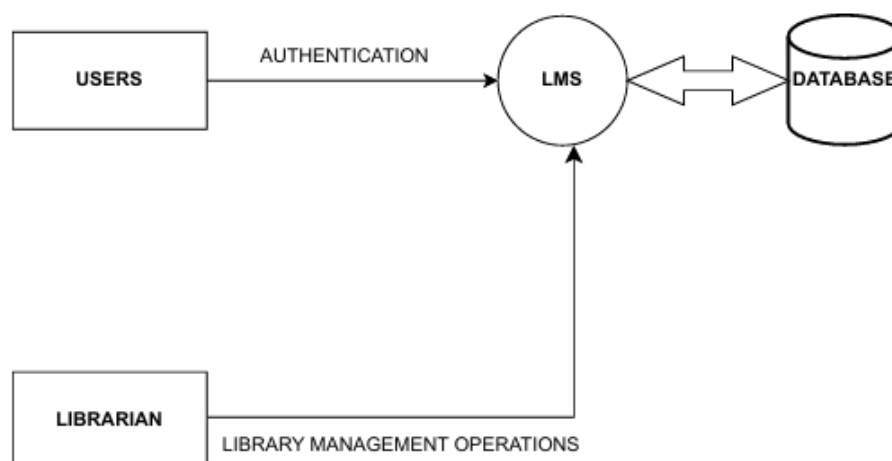


Fig 4.4.1.1 Level-0 DFD

The Fig 4.4.1.1 represents the high-level overview of how data flows between the main components of the system

The diagram represents the high-level overview of the system, showing the interaction between external entities , LMS and Database

MAIN COMPONENTS :

1. USER

- The user interacts with the Library Management System(LMS)
- The user must log in to access the features of the LMS.
- Once authenticated, they can perform various actions like searching for books,borrowing books etc.

2. LIBRARIAN

- The librarian interacts with the system for library management operations, which includes adding,updating, or removing books, issuing books, handling returns, and managing users.

3. LIBRARY MANAGEMENT SYSTEM(LMS)

- Acts as the central processing unit that handles all interactions.
- It verifies user authentication and grants access to authorized users.
- It processes book-related transactions and librarian operations.

4.DATABASE

- Stores user credentials, book records, issued/returned books, and other essential library-related data.
- The LMS retrieves and updates records as needed.

DATA FLOW:

- Users → LMS: Sends authentication requests to log in.
- LMS → Database: Retrieves or verifies user credentials from the database.
- Librarian → LMS: Sends management operations such as book additions,deletions, and updates.
- LMS ↔ Database: Performs data retrieval and updates related to book and user records.

4.4.2 LEVEL-1 DFD

The Level 1 Data Flow Diagram (DFD) provides a more detailed breakdown of the system's processes and interactions

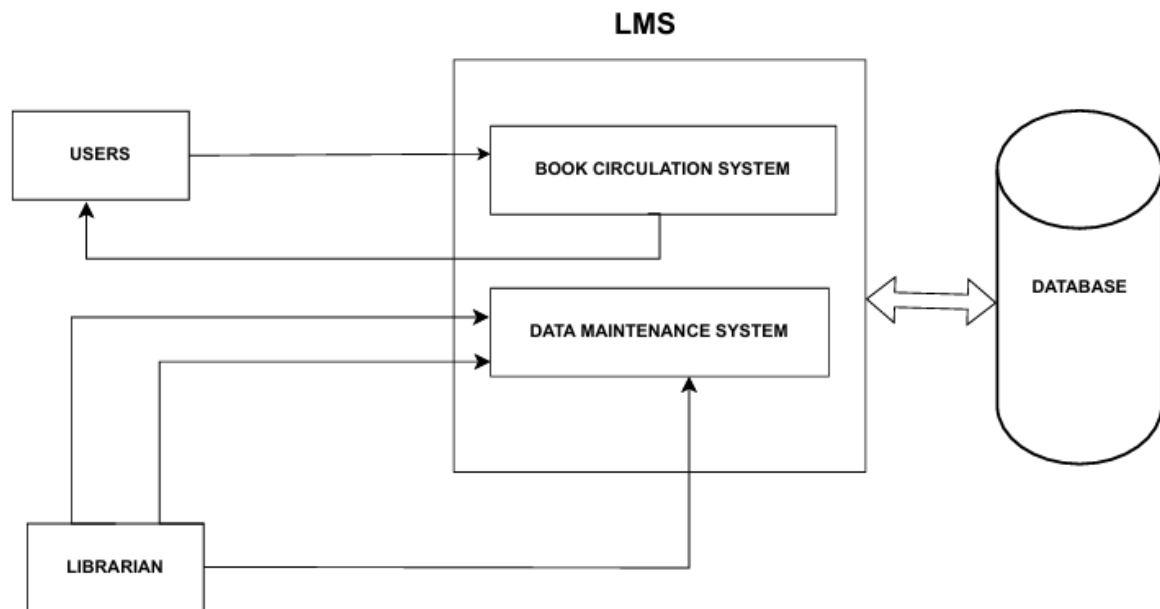


Fig 4.4.2.1 Level-1 DFD

The Level 1 Data Flow Diagram (DFD) as represented by Fig 4.4.2.1 provides a detailed breakdown of the Library Management System (LMS), expanding on the high-level components from Level 0 DFD. It illustrates specific subsystems, data stores, and interactions between the entities (Users, Librarians, and LMS).

MAIN COMPONENTS :

1. BOOK CIRCULATION SYSTEM :

The Book Circulation System is responsible for managing book borrowing and returning processes.

- Authenticates users before granting access.
- Retrieves book details from the Book Info database.
- Verifies member eligibility by checking borrowing history from Member Info.
- Issues and returns books, updating the Loans database accordingly.
- Sends book due dates and return reminders to members.

2. DATA MAINTENANCE SYSTEM :

The Data Maintenance System handles all data updates related to books, members, and transactions.

- Manages book records (add, update, delete books).
- Stores and updates member details, including registration and borrowing history.
- Generates reports for the librarian on book circulation, overdue books, and system usage.
- Communicates with all databases to maintain data integrity.

OVERALL WORKFLOW:

- Users → Book Circulation System: Sends login credentials for authentication.
- Book Circulation System → Database: Retrieves book details based on user requests.
- Book Circulation System → Database: Retrieves member details to verify borrowing eligibility.
- Book Circulation System → Database: Updates loan information when books are issued or returned.
- Librarian → Data Maintenance System: Performs operations like adding books, updating member records, and generating reports.
- Data Maintenance System → Database: Updates loan and member records.
- All Systems ↔ Databases: Continuous data retrieval and updates.

DATABASE:

- Stores book details, availability status, and issue history.
- Stores user credentials and borrowing history.
- Stores information on book issues, due dates, and returns.
- Maintains details of book reservations made by members.

4.4.3 LEVEL-2 DFD

The Level 2 Data Flow Diagram (DFD) provides a detailed breakdown of the internal process within the system. It expands on Level 1 DFD.

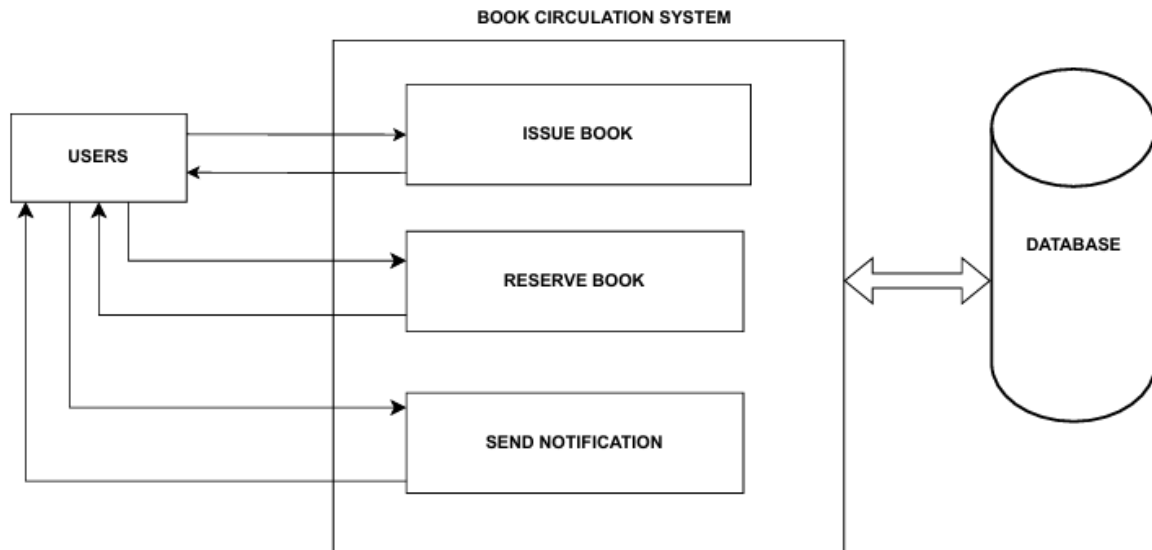


Fig 4.4.3.1 Level-2 DFD 1

The Level 2 DFD 1 as represented by Fig 4.4.3.1 provides a detailed breakdown of the Book Circulation System by expanding its core processes. It highlights how different entities interact with specific subsystems to facilitate book issuance, reservations, and notifications.

MAIN COMPONENTS:

ISSUE BOOK:

- Handles the book lending process.
- Checks user credentials and book availability.
- Updates loan records with due dates.
- Sends book issue details to the Database.

RESERVE BOOK:

- Allows users to reserve books that are currently unavailable.
- Retrieves book and member information from the Database.

- Stores reservation details in the Database for future reference.

SEND NOTIFICATIONS:

- Notifies users about important library events.
- Sends reminders for book return due dates.

DATA FLOW:

- Users → Issue Book: Sends login credentials and book requests.
- Issue Book → Users: Confirms that the book has been issued successfully. Provides the due date and loan details to the user.
- Issue Book → Database: Retrieves book information from the Book Info database.
- Users → Reserve Book: Requests book reservation.
- Reserve Book → Users: If the reserved book becomes available, it is assigned to the user.
- Reserve Book → Database: Checks book availability. Stores reservation details in the Reservation System.
- Users → Send Notification: Users can request notifications for book availability or due date reminders.
- Send Notification → Users: Sends alerts about book due dates. Notifies users when reserved books are available.
- Database → Send Notification: Retrieves information about loan due dates and book reservations.

DATABASE SYSTEM:

- Stores book information, loan details, reservation records, and member details.
- Provides real-time data to different processes for accurate book circulation management.

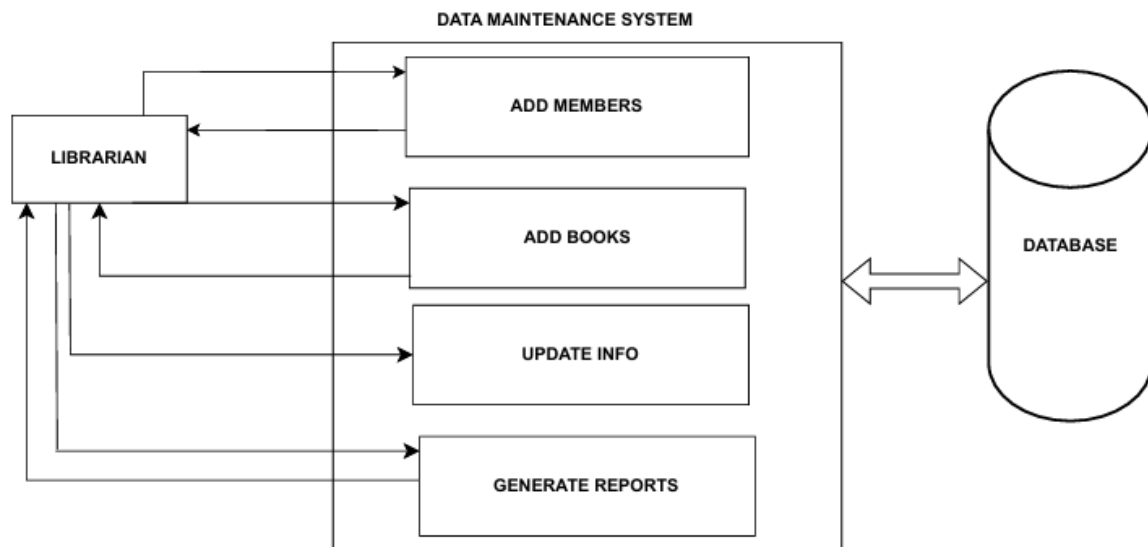


Figure 4.4.3.2 Level-2 DFD 2

Fig 4.3.3.2 shows the Level 2 DFD 2, which represents the Data Maintenance System, responsible for managing and updating library records efficiently. The core components of this system includes Add Members, Add Books, Update Info, Generate Reports.

MAIN COMPONENTS:

ADD MEMBERS:

- The librarian adds new members to the library system.
- Stores member information in the Member Info database.

ADD BOOKS:

- The librarian adds new books to the system.
- Updates the Book Info database with book details (title, author, ISBN, availability, etc.).

UPDATE INFO:

- The librarian updates book/member information as needed.
- Changes are reflected in the database.

GENERATE REPORTS:

- Generates detailed reports on book loans, reservations, and member activity.

- Fetches data from Book Info, Member Info, Loan, and Reservation databases.

DATA FLOW:

- Librarian → Add Members: Provides new member details.
- Add Members → Database: Stores new member information.
- Database → Add Members: Retrieves member records if needed.
- Librarian → Add Books: Enters book details.
- Add Books → Database: Stores book information.
- Database → Add Books: Retrieves book details if required.
- Librarian → Update Info: Requests modification of book/member data.
- Update Info → Database: Updates records.
- Database → Update Info: Retrieves existing information before modification.
- Librarian → Generate Reports: Requests reports on books, members, and transactions.
- Generate Reports → Database: Fetches required data.
- Database → Generate Reports: Supplies necessary information.
- Generate Reports → Librarian: Provides detailed reports.

DATA BASE SYSTEM:

It maintains book records by storing information about books, such as their availability and current status. Additionally, it tracks updates by keeping modification logs for both book and member records. Furthermore, the database generates reports by storing data related to book transactions, member activities, and system updates and this report can be viewed by the admin giving information about book details and issues. Administrators can access these consolidated reports through a dedicated dashboard interface, which provides analytical insights into various metrics such as popular titles, overdue items, peak borrowing periods, and member activity levels. This comprehensive data management approach enables evidence-based decision making while ensuring transparency across all library operations.

4.5 SEQUENCE DIAGRAM

A Sequence Diagram shown by Fig 4.5.1 is a used to visualize the interaction between objects in a sequential order illustrating object interactions, message flows, and the sequence of operations

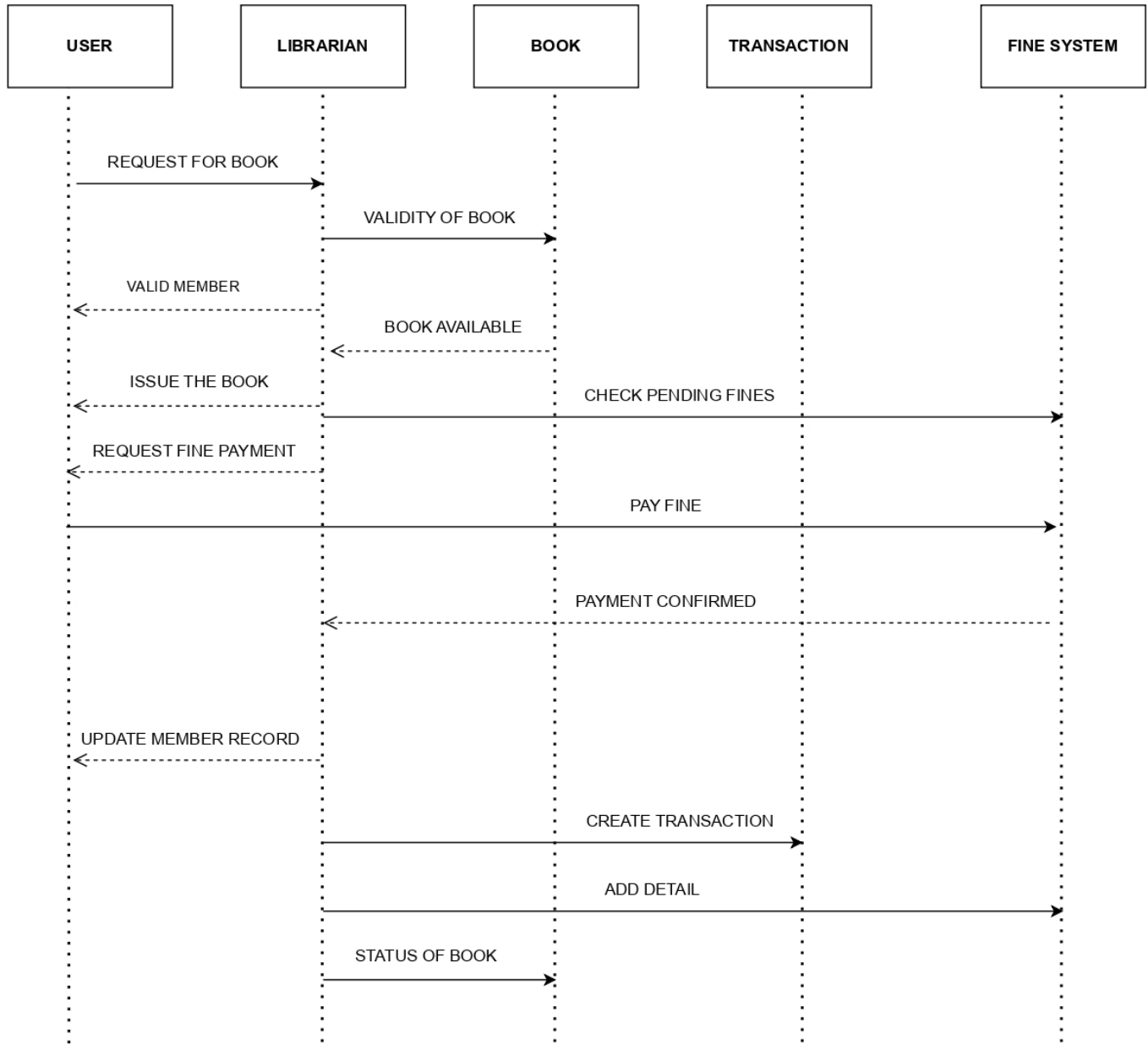


Fig 4.5.1 Sequence Diagram

CHAPTER 5

TESTING AND VALIDATION

5.1 SOFTWARE TESTING OVERVIEW

The Software Testing phase for the Library Management System (LMS) ensures the system meets functional, non-functional, and user-centric requirements while addressing the challenges identified in earlier phases (e.g., scalability, security, and usability). This phase validates the system's reliability, performance, and alignment with the objectives defined in the problem statement. Below is a structured overview of the testing approach:

5.1.1 TESTING OBJECTIVES

- **Functional Validation:** Ensure all core LMS features (book issuance/returns, catalog management, user authentication) operate as specified in the Software Requirements Specification .
- **Performance Assurance:** Verify system responsiveness under peak loads (e.g., concurrent user access during exam periods).
- **Security Compliance:** Protect user data and library records from unauthorized access.
- **Usability Verification:** Validate intuitive UI/UX for librarians and patrons.

5.1.2 KEY TEST SCENARIOS

USER AUTHENTICATION:

- Validate login/logout functionality.
- Test edge cases (invalid credentials).

BOOK MANAGEMENT:

- Verify accurate catalog updates (add/remove books).
- Test search functionality (keyword, author).

TRANSACTION WORKFLOWS:

- Validate book issuance/return processes, including fine calculation.
- Test bulk operations (e.g., issuing multiple books to a user).

REPORTING:

- Ensure accurate generation of reports (e.g., overdue books, user activity).

DIGITAL RESOURCE HANDLING:

- Test upload, download, and access control for e-books/audiobooks.

5.1.3 EXPECTED OUTCOMES

The expected outcome is a LMS that:

- Operates error-free under normal loads.
- Provides seamless access to physical/digital resources.
- Meets user expectations for speed, accuracy, and ease of use.

5.2 TESTING METHODOLOGIES

5.2.1 UNIT TESTING

OBJECTIVE:

- Validate individual components/modules in isolation.

SCOPE:

- Tests for single functions/methods (e.g., fine calculation, search algorithms).
- Mock dependencies (e.g., database calls) for isolated testing.

OUTCOME:

- Ensures each module works correctly before integration.

5.2.2 INTEGRATION TESTING

OBJECTIVE:

- Verify interactions between modules

SCOPE:

- API endpoints (e.g., issue-book interacts with authentication modules).
- Database integration (e.g., book updates reflect in catalogs).

APPROACH:

- Bottom-Up: Test low-level modules first, then higher-level ones.
- Top-Down: Test UI → backend integration early.

5.2.3 PERFORMANCE TESTING

- Evaluated system responsiveness under typical operational conditions
- Simulated multiple concurrent users performing standard library functions: Catalog , Book checkouts and Returns
- Verified database operations with substantial collection sizes
- Monitored and optimized response times for critical functions
- Confirmed system stability under expected workloads
- Ensured efficient report generation capabilities
- Validated bulk transaction processing

5.2.4 SECURITY TESTING

Key areas such as login, password storage, and user roles were tested for vulnerabilities to prevent unauthorized access and data breaches

5.3 TEST CASES

Table 5.3.1 Test Case for LMS

ID	Module	Description	Steps	Expected Result	Status
TC-01	Auth.	Valid Admin Login	1. Enter valid admin credentials 2. Click "Login"	Access granted; dashboard appears	P
TC-02	Auth.	Invalid Login	1. Enter wrong credentials 2. Click "Login"	"Invalid credentials" error	P
TC-03	Books	Add New Book	1. Enter book details 2. Click "Save"	Book appears in catalog	P
TC-04	Books	Search by Keyword	1. Search "Python" 2. Click "Search"	Relevant results shown	P
TC-05	Trans.	Issue Book	1. Search Book 2. Click "Issue"	Book marked "Issued"	P
TC-06	Trans.	Return Overdue	1. Mark overdue book	Fine calculated; book available	P
TC-07	Reports	Overdue Report	1. Select report 2. Generate	Graph with overdue books	P
TC-08	Digital	Upload E-Book	1. Upload PDF 2. Add metadata	E-book appears in library	P
TC-09	Data	Delete Book	1. Delete book 2. Check records	Clean removal	P
TC-10	Data	Delete E-Book	1. Delete E-Book 2. Check records	Clean removal	P
TC-11	User Mgmt	Add New User	1. Admin logs in. 2. Navigates to "Manage User". 3. Click "Add New User"	User added; confirmation message	P
TC-12	User Mgmt	Delete User	1. Admin logs in. 2. Navigates to "Manage User". 3. Click "Delete User"	User Deleted; confirmation message	P

5.4 TESTING TECHNOLOGIES USED

To perform rigorous testing for LMS, the following tools and technologies were used:

- **Manual Testing:** The team manually tested core features like form submissions and navigation. They verified data handling, error messages, and basic workflows.
- **API Testing:** Backend APIs were tested for authentication, catalog operations, and transactions. Testers checked responses, status codes, and data integrity.
- **Black Box Testing:** Testers evaluated user features without internal knowledge. They validated system behavior using various inputs and scenarios.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 OUTPUT SCREENS

6.1.1 LOGIN PAGE

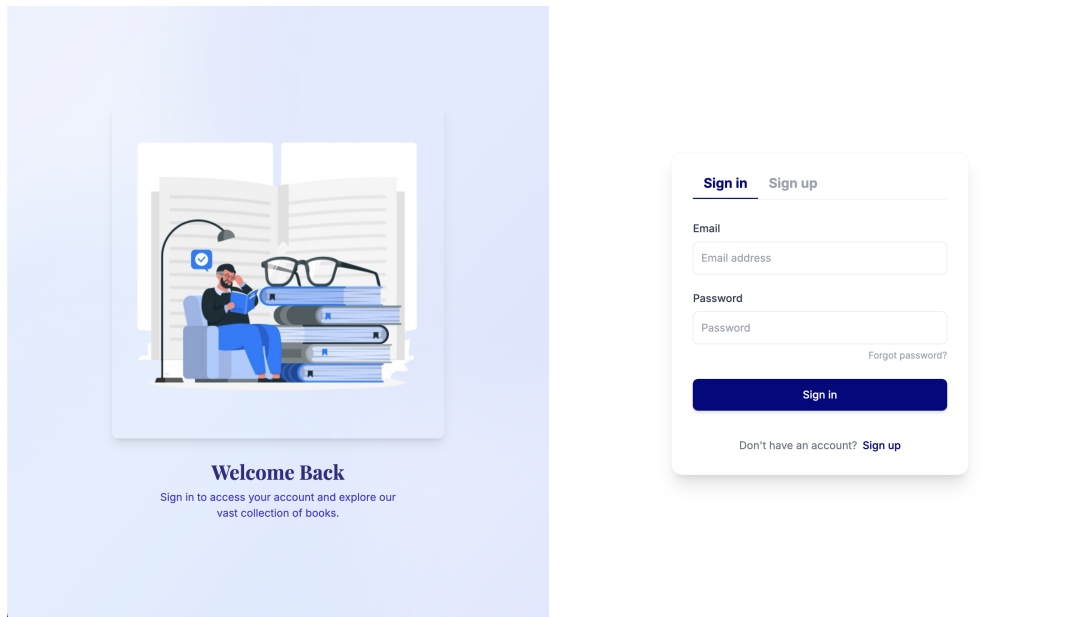


Fig 6.1.1.1 Login Page

6.1.2 REGISTER PAGE

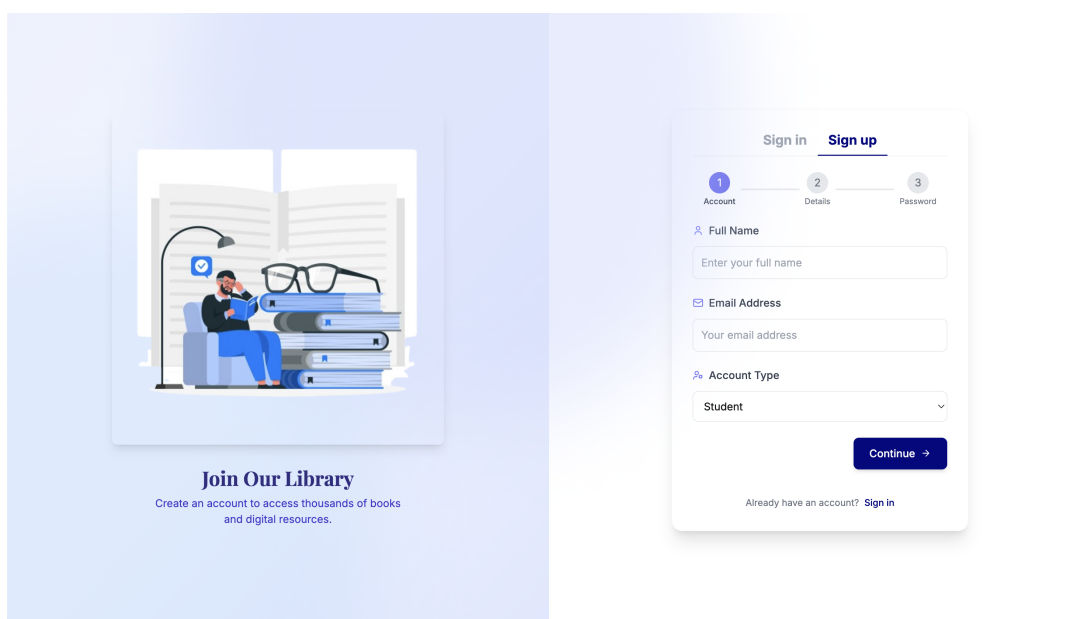


Fig 6.1.2.1 Register Page

6.1.3 DASHBOARD

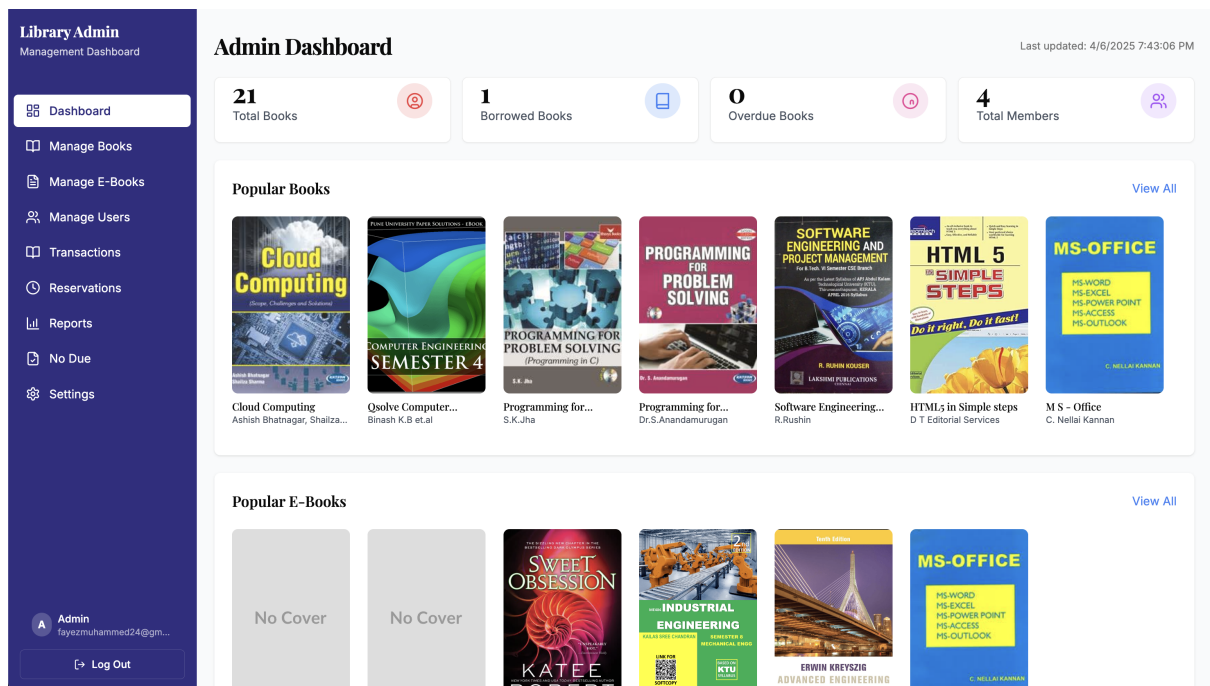


Fig 6.1.3.1 Dashboard

6.1.4 NO-DUE PAGE

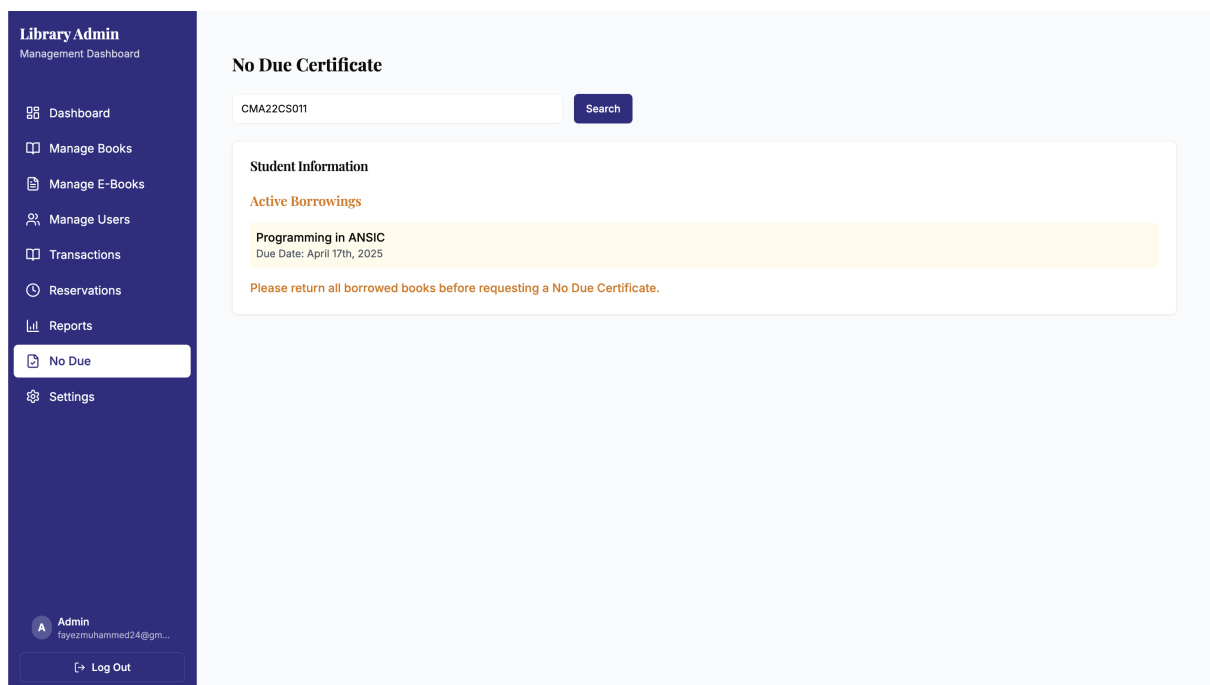


Fig 6.1.4.1 No-Due Page

6.1.5 ADD BOOK PAGE

Library Admin
Management Dashboard

Dashboard | **Manage Books** | Manage E-Books | Manage Users | Transactions | Reservations | Reports | No Due | Settings

Admin
fayezmuhammed24@gm...
Log Out

Add New Book

Required Fields

Book Number *
Enter unique book number

Title *
Enter book title

Author *
Enter author name

Optional Fields

ISBN
Enter ISBN number (e.g., 9781234567890)

Category
Select a category

Status
Available

Copies
1

Publisher

Edition

Description

Tags
Comma-separated tags

Cover Image

Back to Books

Fig 6.1.5.1 Add Book page

6.1.6 MANAGE BOOK PAGE

Library Admin
Management Dashboard

Dashboard | **Manage Books** | Manage E-Books | Manage Users | Transactions | Reservations | Reports | No Due | Settings

Admin
fayezmuhammed24@gm...
Log Out

Manage Books

Search by title, author, or ISBN...

+ Add New Book

Book No.	Title	Author	ISBN	Category	Status	Copies	Actions
CSE/19/1	Cloud Computing	Ashish Bhatnagar, Shailza Sharma		Book	Available	1	Edit Delete
CSE/19/2	Qsolve Computer Science & Engineering Semeter 4 (2)	Binash K.B et.al		Book	Available	1	Edit Delete
CSE/19/3	Programming for problem solving	S.K.Jha		Science	Available	2	Edit Delete
CSE/19/4	Programming for problem solving	Dr.S.Anandamurugan		Text Books	Available	1	Edit Delete
CSE/19/5	Software Engineering and Project Management	R.Rushin		Text Books	Available	1	Edit Delete
CSE/19/6	HTML5 in Simple steps	D T Editorial Services		Book	Available	1	Edit Delete
CSE/19/7	M S - Office	C. Nellai Kannan		Text Books	Available	1	Edit Delete
CSE/19/8	Test Your C skills	Yashavant kanetkar		Book	Available	1	Edit Delete
CSE/19/9	Introduction To Automation Theory, Languages & Computation	John E.Hopcroft, Jeffery D.Ullman		Text Books	Available	1	Edit Delete
CSE/19/10	Data Structures Through C	G.S.Baluja		Text Books	Issued	1	Edit Delete

Fig 6.1.6.1 Manage Book Page

CHAPTER 7

CONCLUSION

This Library Management System project has developed a practical solution to modernize and simplify library operations. By focusing on user needs and operational efficiency, we've created a system that effectively manages book cataloging, circulation, and reporting processes. The software replaces error-prone manual methods with reliable digital workflows, enabling librarians to serve patrons more effectively while reducing administrative burdens. Key features like the intuitive search interface and automated record-keeping demonstrate how thoughtful design can enhance both staff productivity and user satisfaction. While implementing the system, we addressed challenges related to data migration and workflow adaptation, ensuring the transition from traditional methods would be smooth for libraries of various sizes. The project successfully balances comprehensive functionality with straightforward operation, making advanced library management accessible without requiring technical expertise. Looking ahead, there are opportunities to expand the system's capabilities, particularly in supporting mobile access and additional digital media formats. This work highlights how targeted technological solutions can help libraries maintain their vital role as information centers while adapting to evolving user expectations. The system provides a solid foundation that libraries can build upon as their needs grow and change in the coming years.

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APPENDICES

RESERVATION MANAGEMENT

```
const Reservation = require('../models/Reservation');
const Book = require('../models/Book');
const ErrorResponse = require('../utils/errorResponse');
const asyncHandler = require('../middleware/async');
// @desc Reserve a book
// @route POST /api/reservations
// @access Private
exports.reserveBook = asyncHandler(async (req, res, next) => {
  const { bookId } = req.body;
  // Check if book exists
  const book = await Book.findById(bookId);
  if (!book) {
    return next(new ErrorResponse('Book not found with id of ${bookId}', 404));
  }
  // Check if book is available
  if (book.status !== 'Available') {
    return next(new ErrorResponse('Book is not available for reservation', 400));
  }
  // Check if user already has an active reservation for this book
  const existingReservation = await Reservation.findOne({
    user: req.user.id,
    book: bookId,
    status: 'active'
  });
  if (existingReservation) {
    return next(new ErrorResponse('You already have an active reservation for this
    book', 400));
  }
  // Create reservation
  const reservation = await Reservation.create({
    user: req.user.id,
    book: bookId
  });
  await Book.findByIdAndUpdate(bookId, { status: 'Reserved' });
});
```

```

        res.status(201).json({
            success: true,
            message: 'Book reserved successfully. Your reservation is valid for 24 hours.',
            data: reservation
        });
    });
// @desc   Get user's reservations
// @route  GET /api/reservations
// @access Private
exports.getUserReservations = asyncHandler(async (req, res, next) => {
    const reservations = await Reservation.find({
        user: req.user.id,
        status: 'active'
    })
    .populate({
        path: 'book',
        select: 'title author isbn status coverImage image imagePath'
    })
    .sort({ reservedAt: -1 });

    res.status(200).json({
        success: true,
        count: reservations.length,
        data: reservations
    });
});
exports.cancelReservation = asyncHandler(async (req, res, next) => {
    let reservation = await Reservation.findById(req.params.id);

    if (!reservation) {
        return next(new ErrorResponse('Reservation not found with id of ${req.params.id}', 404));
    }
    // Make sure user is reservation owner
    if (reservation.user.toString() !== req.user.id && req.user.role !== 'admin') {
        return next(new ErrorResponse('User not authorized to cancel this reservation', 401));
    }
});

```

```
// Update reservation status
reservation.status = 'cancelled';
await reservation.save();

// Update book status back to 'Available'
await Book.findByIdAndUpdate(reservation.book, { status: 'Available' });

res.status(200).json({
  success: true,
  message: 'Reservation cancelled successfully',
  data: {}
});
});

exports.getAllReservations = asyncHandler(async (req, res, next) => {
  const reservations = await Reservation.find()
    .populate({
      path: 'book',
      select: 'title author isbn status'
    })
    .populate({
      path: 'user',
      select: 'name email'
    })
    .sort({ reservedAt: -1 });

  res.status(200).json({
    success: true,
    count: reservations.length,
    data: reservations
  });
});

exports.expireOutdatedReservations = asyncHandler(async (req, res, next) => {
  const now = new Date();
  const expiredReservations = await Reservation.find({
    status: 'active',
    expiresAt: { $lt: now }
  });
});
```

```
// Update reservation statuses and book statuses
const updates = expiredReservations.map(async (reservation) => {
  // Update reservation status
  reservation.status = 'expired';
  await reservation.save();

  // Update book status
  await Book.findByIdAndUpdate(reservation.book, { status: 'Available' });

  return reservation;
});

await Promise.all(updates);

res.status(200).json({
  success: true,
  count: expiredReservations.length,
  message: `${expiredReservations.length} outdated reservations expired
    successfully`,
  data: expiredReservations
});
});
```